

Catalytic pyrolysis of LDPE leads to valuable resource recovery and reduction of waste problems

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ABSTRACT

Recycling of waste polymers has become a necessity because huge piles of those polymers represent a threat to the environment. Used polymers are also a source of energy and valuable chemicals.

Used low density polyethylenes (LDPE) were catalytically pyrolysed in a home assembled batch reactor under atmospheric pressure. For maximum conversion into chemicals which could be used for feedstock recovery optimum conditions like temperature, catalyst weight and reaction time were optimized. A wide range of acidic and basic catalysts like silica, calcium carbide, alumina, magnesium oxide, zinc oxide and homogeneous mixture of silica and alumina were tried for this purpose.

Though CaC_2 was better on the basis of reaction time, however the efficiency of conversion into liquid for SiO_2 was found to be maximum at optimum conditions. These two catalysts could be picked up as suitable catalysts for catalytic pyrolysis of polyethylene. The results of the column separation using different solvents indicate that the oxide containing catalyst could be best suited for selective conversion into polar and aromatic products while CaC_2 catalyst could be adopted for selective conversion into aliphatic products.

The liquid product obtained from catalytic pyrolysis was also characterized by physical and chemical tests. Among the physical tests density, specific gravity, API gravity, viscosity, kinematic viscosity, aniline point, flash point, Watson characterization constant, freezing point, diesel index, refractive index, gross calorific value, Net calorific value and ASTM Distillation were determined according to IP and ASTM standard methods for fuel values. From the physical tests it was observed that the results for the liquid fractions are comparable with the standard results of physical tests for gasoline, kerosene and diesel fuel oil. From the Bromine water and KMnO_4 tests it was observed that liquid obtained is a mixture of olefin and aromatic hydrocarbons. This was further confirmed by Bromine number tests. The values of which lie in the range of 0.1–12.8 g/ml, which fall in the range for olefin mixture. Phenol and carbonyl contents were quantified using UV/Visible spectroscopy and the values lie in the range of 1–8920 $\mu\text{g/ml}$ and 5–169 $\mu\text{g/ml}$ for both phenols and carbonyls respectively. The components of different hydrocarbons in the oil mixture were separated by using column chromatography and fractional distillation followed by characterization with FT-IR spectroscopy.

The interpretation of FT-IR spectra shows that catalytic pyrolysis of LDPE leads to the formation of a complex mixture of alkanes, alkenes, carbonyl group containing compounds like aldehydes, ketones, aromatic compounds and substituted aromatic compounds like phenols. It could be concluded, that catalytic pyrolysis of LDPE leads to valuable resource recovery and reduction of waste problem.

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1. Introduction

Used Low density polyethylene from household and industries are recognized to be a major environmental problem. There are several methods for disposal of municipal and industrial LDPE wastes, i.e. landfill, incineration, true material recycling, and chemical recovery. Landfill treatment and incineration destruction are

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mixture was refluxed for 2 h. The refluxed oil was characterized using physical and chemical tests given below.

2.2.2. Determination of physiochemical properties of the liquid product

The liquid column chromatography was used to separate the different groups of hydrocarbons present in the derived oil. The silica gel 60 (63–200 μm grain size, supplied by Merck) was packed into borosilicate glass column and the pyrolysis oil applied to the top of the column. The column was then eluted with n-hexane, benzene and methanol to produce aliphatic, aromatic and polar fractions of the pyrolysis oil, respectively.

Fractional distillation was carried out to separate different boiling point hydrocarbon fractions. The oil was distilled at a specific temperature until no more distilled products were collected.

Infrared spectroscopy was used for identification of liquid product measuring absorption from 400 to 4400 cm^{-1} .

Phenols and carbonyls were quantitatively determined by spectrophotometric methods using Folin–Denis and phenyl hydrazine reagents respectively.

The fractions obtained by this method were also characterized by using density, refractive index, refractive index parameter (I) and refractivity intercepts (RI). The refractive indexes were calculated with Abbe's refractometer (P20-Warsaw, Poland) at 20 °C. Refractive index parameter and refractivity intercepts were calculated for all the fractions by using the formulas:

$$\text{Refractive index parameter (I)} = \eta^2 - 1/\eta^2 + 2$$

$$\text{Refractivity Intercept (RI)} = \eta - d/2$$

η = Refractive index, d = density.

Some physical properties of pyrolytic oils were determined by using the following standard methods: Flash point by Cleveland open cup method IP-36/84 and ASTM-D92-78. Density IP-59/82, API gravity IP-160/87 and ASTM-D1298-85, kinematic viscosity IP-711/87 and ASTM-D445-87 and distillation IP-191/83 and ASTM-D216-77 were determined according to IP and ASTM standard methods for fuel.

Gross calorific values were determined using Gallenkamp Auto Bomb Calorimeter (England).

3. Results and discussions of polyethylene

3.1. Thermogravimetric analysis of polyethylene

The weight loss versus temperature curve for thermal decomposition of polyethylene under a linear heating rate (Fig. 1) shows that the decomposition was a single step process with offset and end temperature of 175.5 °C and 400 °C respectively. The curve shows that maximum weight loss related to volatilization of hydrocarbons occurred at 400 °C. Total 99.98% weight changes occurred from 175.5 °C to 400 °C. The residue weight was 0.2% and was found constant up to 1000 °C. This indicates that certain amount of char like carbeneous material is formed during the decomposition of polyethylene. Along with the volatilization process, other reactions like cracking of side chains from aromatic rings, isomerization and polycondensation also occur. Therefore it indicates that all volatile matter of polyethylene has been decomposed to volatile hydrocarbons.

3.2. Optimization of temperature for pyrolysis of polyethylene

Polyethylene sample was pyrolysed at different temperatures. The results are given in Table 1. It can be seen from these results that maximum conversion was obtained at 400 °C. At this temperature conversion into liquid oil product was $12.0 \pm 0.2\%$, while conversion into gaseous product was $13.36 \pm 0.32\%$. Total conversion at this temperature was $25.36 \pm 0.52\%$.

3.3. Optimization study for catalytic pyrolysis of polyethylene

Polyethylene sample was catalytically pyrolysed at varied temperatures using different catalysts like ZnO, MgO, CaC_2 , SiO_2 , Al_2O_3 and mixture of SiO_2 and Al_2O_3 . The results are given in Tables 2 and 3. The product in all cases was liquid oil, gases and solid residue. The purpose of temperature optimization was to find out suitable temperature at which maximum liquid product could be obtained. Total conversion was based on the amount of liquid and gas formed during catalytic pyrolysis. It can be seen from the results that maximum conversion into liquid product for MgO,

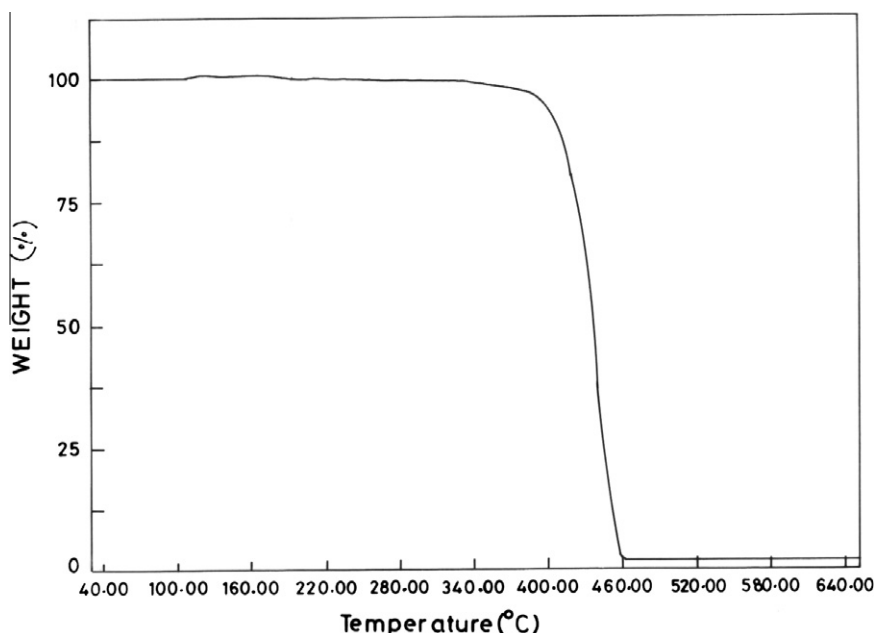


Fig. 1. TGA curve of polyethylene.

Table 1
Optimum temperature investigation for thermal pyrolysis of polyethylene.

S. no.	Temp. (°C)	$\bar{x}$			
		L $\pm$ S.D	S $\pm$ S.D	G $\pm$ S.D	T. Con. L + G $\pm$ S.D
1	250	2.26 $\pm$ 0.25	89.43 $\pm$ 0.11	8.26 $\pm$ 0.20	10.52 $\pm$ 0.45
2	300	5.10 $\pm$ 0.10	84.13 $\pm$ 0.34	10.76 $\pm$ 0.25	15.86 $\pm$ 0.35
3	350	10.0 $\pm$ 0.20	72.36 $\pm$ 0.12	12.36 $\pm$ 0.32	22.36 $\pm$ 0.52
4	400	12.0 $\pm$ 0.20	66.43 $\pm$ 0.93	13.36 $\pm$ 0.32	25.36 $\pm$ 0.52

Conditions: Weight of polyethylene sample = 5.0 g, Temp. = Temperature, S.D = Standard deviation.

Heating time = 60 min, L = liquid, S = solid, G = gas, T. Con. L + G (%) = total percent conversion into liquid and gas, where $\bar{x}$ = Average.

CaC₂, SiO₂ and Al₂O₃ takes place at 350 °C, while in case of ZnO and mixture of SiO₂ and Al₂O₃ optimum temperature was found to be 400 °C. Beyond the optimum temperature a decrease in the liquid fraction was observed which could be due to further conversion into volatile product leading to higher fraction of gases and decrease in liquid fraction. Comparing the catalytic pyrolysis ability of those catalysts it can be seen that maximum liquid product could be obtained using CaC₂ as a catalyst. Further maximum conversion of 98.76% was also achieved with this catalyst. The efficiency of conversion into liquid fraction for ZnO and mixture of SiO₂ and Al₂O₃ was the same and was found to be next to that of CaC₂. Among the catalysts investigated CaC₂ was found to be most suitable for catalytic pyrolysis of polyethylene.

In the next step the effect of catalyst weight on the catalytic pyrolysis of polyethylene for these catalysts was investigated in

the range of 0–2.5 g for 5 g of LDPE sample at optimum temperature. The results are given in Tables 4 and 5. It was observed that 1 g of catalyst weight was found as optimum weight for 5 g of LDPE sample in case of SiO₂ and CaC₂, 1.5 g for MgO and Al₂O₃ and 2 g for ZnO and equimolar mixture of SiO₂ and Al₂O₃ as catalyst. Below the optimum catalyst weight conversion into liquid oil was minimum, while higher weight of catalyst lead to higher gaseous product. This is due to the fact that at higher catalyst weight the rate of reaction increases the cracking process, which results in maximum gaseous product.

The effect of residence time on the catalytic pyrolysis was also studied at optimum conditions of temperature and catalyst weight and the results are given in Tables 6 and 7a. It was observed that 60 min reaction time was sufficient for maximum conversion in case of CaC₂, while in case of MgO and Al₂O₃ optimum reaction time was found to be 120 min. The reaction was slow in case of ZnO, SiO₂ and mixture of SiO₂ and Al₂O₃ and required 180 min for maximum conversion.

Maximum conversion at optimum condition of temperature, catalyst weight and reaction time for ZnO, MgO, CaC₂, SiO₂, Al₂O₃ and mixture of SiO₂ and Al₂O₃ was found to be 64.93 $\pm$ 0.986, 66.23 $\pm$ 0.251, 69.73 $\pm$ 0.64, 79.10 $\pm$ 0.71, 62.23 $\pm$ 0.25 and 67.43 $\pm$ 0.208 respectively. The comparative study of all these catalysts at optimum conditions is shown in Table 7b.

Though CaC₂ was better on the basis of reaction time, however the efficiency of conversion into liquid for SiO₂ was found to be maximum at optimum conditions. These two catalysts could be picked up as suitable catalysts for catalytic pyrolysis of polyethylene.

Table 2
Optimum temperatures investigation for catalytic pyrolysis of polyethylene using different catalysts (wt%).

Type of catalyst	Temp. (°C)	$\bar{x}$			
		L $\pm$ S.D	S $\pm$ S.D	G $\pm$ S.D	T. Con. L + G $\pm$ S.D
ZnO	250	0.00 $\pm$ 0.00	95.56 $\pm$ 0.152	3.43 $\pm$ 0.152	3.43 $\pm$ 0.152
	300	20.3 $\pm$ 0.02	73.4 $\pm$ 0.556	6.3 $\pm$ 0.360	26.6 $\pm$ 0.556
	350	30.9 $\pm$ 0.360	54.7 $\pm$ 0.624	14.4 $\pm$ 0.264	45.3 $\pm$ 0.624
	400	33.7 $\pm$ 0.264	39.7 $\pm$ 0.435	20.16 $\pm$ 0.152	60.30 $\pm$ 0.435
MgO	250	0.0 $\pm$ 0.00	97.76 $\pm$ 0.251	2.23 $\pm$ 0.25	2.23 $\pm$ 0.251
	300	18.3 $\pm$ 0.264	72.3 $\pm$ 0.458	9.0 $\pm$ 0.2	27.7 $\pm$ 0.458
	350	32.76 $\pm$ 0.251	51.86 $\pm$ 0.416	15.36 $\pm$ 0.32	48.13 $\pm$ 0.416
	400	31.53 $\pm$ 0.416	47.9 $\pm$ 1.80	18.6 $\pm$ 0.14	52.76 $\pm$ 0.763
CaC ₂	250	25.3 $\pm$ 0.608	72.36 $\pm$ 0.55	2.33 $\pm$ 0.57	27.63 $\pm$ 0.550
	300	51.5 $\pm$ 0.50	28.76 $\pm$ 0.404	19.73 $\pm$ 0.64	71.23 $\pm$ 0.404
	350	69.73 $\pm$ 0.64	1.43 $\pm$ 0.814	28.83 $\pm$ 0.76	98.56 $\pm$ 0.814
	400	51.0 $\pm$ 1.0	1.23 $\pm$ 0.68	33.33 $\pm$ 0.57	98.76 $\pm$ 0.680

Table 3
Optimum temperatures investigation for catalytic pyrolysis of polyethylene using different catalysts (wt%).

Type of catalyst	Temp. (°C)	$\bar{x}$			
		L $\pm$ S.D	S $\pm$ S.D	G $\pm$ S.D	T. Con. L + G $\pm$ S.D
SiO ₂	250	0.0 $\pm$ 0.000	95.8 $\pm$ 0.264	4.20 $\pm$ 0.200	4.20 $\pm$ 0.200
	300	24.2 $\pm$ 0.264	68.2 $\pm$ 0.264	7.6 $\pm$ 0.400	31.8 $\pm$ 0.264
	350	33.56 $\pm$ 0.602	50.93 $\pm$ 0.92	15.5 $\pm$ 0.500	49.06 $\pm$ 0.929
	400	30.83 $\pm$ 0.763	43.0 $\pm$ 1.73	20.5 $\pm$ 0.500	51.6 $\pm$ 1.269
Al ₂ O ₃	250	0.0 $\pm$ 0.0	95.53 $\pm$ 0.152	3.46 $\pm$ 0.152	3.46 $\pm$ 0.153
	300	21.5 $\pm$ 0.50	72.63 $\pm$ 0.152	6.2 $\pm$ 0.351	27.36 $\pm$ 0.152
	350	31.43 $\pm$ 0.493	55.16 $\pm$ 0.115	13.4 $\pm$ 0.458	44.83 $\pm$ 0.115
	400	29.4 $\pm$ 0.360	44.26 $\pm$ 0.873	19.1 $\pm$ 0.36	55.73 $\pm$ 0.873
SiO ₂ –Al ₂ O ₃	250	0.0 $\pm$ 0.00	95.86 $\pm$ 0.152	4.13 $\pm$ 0.152	4.13 $\pm$ 0.152
	300	19.23 $\pm$ 0.251	74.53 $\pm$ 0.461	6.23 $\pm$ 0.251	25.46 $\pm$ 0.461
	350	29.26 $\pm$ 0.251	56.13 $\pm$ 0.513	14.6 $\pm$ 0.360	43.86 $\pm$ 0.513
	400	34.16 $\pm$ 0.763	41.43 $\pm$ 0.404	21.16 $\pm$ 0.288	58.56 $\pm$ 0.404

Table 4

Optimum catalyst weight investigation for catalytic pyrolysis of polyethylene using different catalysts (wt%).

Type of catalyst	Catalyst/feed ratio	$\bar{x}$			
		L $\pm$ S.D	S $\pm$ S.D	G $\pm$ S.D	T. Con. L + G $\pm$ S.D
ZnO	0	12.0 $\pm$ 0.2	66.43 $\pm$ 0.929	13.36 $\pm$ 0.321	25.36 $\pm$ 0.521
	0.1	24.46 $\pm$ 0.152	52.06 $\pm$ 0.602	24.46 $\pm$ 0.152	55.9 $\pm$ 0.264
	0.2	33.7 $\pm$ 0.264	39.7 $\pm$ 0.435	20.16 $\pm$ 0.152	60.30 $\pm$ 0.435
	0.3	35.3 $\pm$ 0.30	37.16 $\pm$ 0.057	35.3 $\pm$ 0.30	76 $\pm$ 0.721
	0.4	38.4 $\pm$ 0.529	37.10 $\pm$ 0.173	38.4 $\pm$ 0.529	76.8 $\pm$ 1.05
MgO	0.5	37.63 $\pm$ 0.152	36.06 $\pm$ 0.288	37.63 $\pm$ 0.152	75.26 $\pm$ 0.305
	0	10.0 $\pm$ 0.2	72.36 $\pm$ 0.115	12.36 $\pm$ 0.321	22.36 $\pm$ 0.521
	0.1	26.73 $\pm$ 0.25	55.93 $\pm$ 0.305	14.1 $\pm$ 0.10	44.06 $\pm$ 0.305
	0.2	32.76 $\pm$ 0.251	51.86 $\pm$ 0.416	15.36 $\pm$ 0.32	48.13 $\pm$ 0.416
	0.3	40.2 $\pm$ 0.1	36.1 $\pm$ 0.360	23.7 $\pm$ 0.264	63.9 $\pm$ 0.360
CaC ₂	0.4	39.86 $\pm$ 0.321	35.06 $\pm$ 0.378	25.06 $\pm$ 0.115	64.93 $\pm$ 0.378
	0	10.0 $\pm$ 0.20	72.36 $\pm$ 0.115	12.36 $\pm$ 0.321	22.36 $\pm$ 0.521
	0.1	29.0 $\pm$ 0.2	46.0 $\pm$ 0.435	20.76 $\pm$ 0.251	54.0 $\pm$ 0.435
	0.2	69.73 $\pm$ 0.64	1.43 $\pm$ 0.814	28.83 $\pm$ 0.76	98.56 $\pm$ 0.814
	0.3	66.86 $\pm$ 0.31	0.70 $\pm$ 0.30	32.43 $\pm$ 0.404	99.30 $\pm$ 0.300

Table 5

Optimum catalyst weights Investigation for catalytic pyrolysis of polyethylene using different catalysts (wt%).

Type of catalyst	Catalyst/feed ratio	$\bar{x}$			
		L $\pm$ S.D	S $\pm$ S.D	G $\pm$ S.D	T. Con. L + G $\pm$ S.D
SiO ₂	0.1	21.26 $\pm$ 0.251	64.1 $\pm$ 0.000	13.3 $\pm$ 0.282	35.9 $\pm$ 0.00
	0.2	33.56 $\pm$ 0.602	50.93 $\pm$ 0.92	15.5 $\pm$ 0.500	49.06 $\pm$ 0.929
	0.3	32.36 $\pm$ 0.321	50.43 $\pm$ 0.115	17.2 $\pm$ 0.264	49.56 $\pm$ 0.11
Al ₂ O ₃	0	10.0 $\pm$ 0.20	72.36 $\pm$ 0.115	12.36 $\pm$ 0.321	22.36 $\pm$ 0.521
	0.1	20.23 $\pm$ 0.321	64.53 $\pm$ 0.321	12.96 $\pm$ 0.057	33.19 $\pm$ 0.378
	0.2	31.43 $\pm$ 0.493	55.16 $\pm$ 0.115	13.4 $\pm$ 0.458	44.83 $\pm$ 0.115
	0.3	35.8 $\pm$ 0.2	49.76 $\pm$ 0.461	14.43 $\pm$ 0.305	50.28 $\pm$ 0.505
	0.4	35.03 $\pm$ 0.529	48.63 $\pm$ 0.321	16.23 $\pm$ 0.251	51.26 $\pm$ 0.78
SiO ₂ –Al ₂ O ₃	0	12.0 $\pm$ 0.200	66.43 $\pm$ 0.929	13.36 $\pm$ 0.321	25.36 $\pm$ 0.521
	0.1	25.16 $\pm$ 0.288	53.83 $\pm$ 0.152	15.7 $\pm$ 0.264	46.13 $\pm$ 0.152
	0.2	34.16 $\pm$ 0.763	41.43 $\pm$ 0.404	21.16 $\pm$ 0.288	58.56 $\pm$ 0.404
	0.3	37.33 $\pm$ 0.288	38.5 $\pm$ 0.458	22.3 $\pm$ 0.200	61.5 $\pm$ 0.458
	0.4	41.23 $\pm$ 0.251	35.83 $\pm$ 0.458	22.93 $\pm$ 0.115	64.16 $\pm$ 0.351
	0.5	40.33 $\pm$ 0.577	35.23 $\pm$ 0.251	24.43 $\pm$ 0.404	64.76 $\pm$ 0.251

Table 6

Optimum heating time investigation for catalytic pyrolysis of polyethylene using different catalysts (wt%).

Type of catalyst	Time (min)	$\bar{x}$			
		L $\pm$ S.D	S $\pm$ S.D	G $\pm$ S.D	T. Con. L + G $\pm$ S.D
ZnO	30	19.6 $\pm$ 0.321	68.43 $\pm$ 0.351	12.2 $\pm$ 0.20	31.56 $\pm$ 0.351
	60	33.7 $\pm$ 0.264	39.7 $\pm$ 0.435	20.16 $\pm$ 0.152	60.30 $\pm$ 0.435
	90	44.03 $\pm$ 0.55	27.56 $\pm$ 0.901	28.4 $\pm$ 0.360	72.43 $\pm$ 0.901
	120	50.36 $\pm$ 0.152	19.4 $\pm$ 0.360	30.23 $\pm$ 0.321	80.6 $\pm$ 0.360
	150	60.56 $\pm$ 0.404	6.66 $\pm$ 0.568	32.76 $\pm$ 0.251	93.33 $\pm$ 0.568
	180	64.93 $\pm$ 0.986	0.83 $\pm$ 1.184	34.23 $\pm$ 0.251	99.16 $\pm$ 1.184
MgO	30	22.43 $\pm$ 0.208	63.4 $\pm$ 0.173	14.16 $\pm$ 0.152	36.6 $\pm$ 0.173
	60	32.76 $\pm$ 0.251	51.86 $\pm$ 0.416	15.36 $\pm$ 0.32	48.13 $\pm$ 0.416
	90	49.86 $\pm$ 0.321	20.9 $\pm$ 0.4	29.23 $\pm$ 0.251	79.1 $\pm$ 0.400
	120	66.23 $\pm$ 0.251	0.4 $\pm$ 0.264	33.36 $\pm$ 0.152	99.6 $\pm$ 0.264
CaC ₂	30	10.50 $\pm$ 0.5	76.26 $\pm$ 0.25	13.23 $\pm$ 0.64	23.75 $\pm$ 0.125
	60	69.73 $\pm$ 0.64	1.43 $\pm$ 0.814	28.83 $\pm$ 0.76	98.56 $\pm$ 0.814
	90	62.23 $\pm$ 0.25	1.33 $\pm$ 0.72	36.43 $\pm$ 0.92	98.66 $\pm$ 1.171

3.4. Fractionation and separation of derived oil using column chromatography

Separation of oil derived by catalytic pyrolysis of polyethylene was carried out on silica gel column. The sample was loaded onto the column and elution was done using different solvents like

n-hexane, benzene and methanol. It was expected that aliphatic fraction will be eluted with n-hexane, aromatic with benzene and polar hydrocarbons with methanol. The results are given in Table 8. It can be seen from the results that when SiO₂, Al₂O₃, MgO and ZnO is used as a catalyst, the derived oil has maximum amount of polar fraction followed by non-polar and relatively polar. In case

Table 7a

Optimum heating time investigation for catalytic pyrolysis of polyethylene using different catalysts (wt%).

Type of catalyst	Time (min)	$\bar{x}$			
		L $\pm$ S.D	S $\pm$ S.D	G $\pm$ S.D	T. Con. L + G $\pm$ S.D
SiO ₂	30	10.26 $\pm$ 0.208	84.0 $\pm$ 0.781	5.73 $\pm$ 0.642	16.0 $\pm$ 0.781
	60	33.56 $\pm$ 0.602	50.93 $\pm$ 0.92	15.5 $\pm$ 0.500	49.06 $\pm$ 0.929
	90	52.1 $\pm$ 0.360	31.03 $\pm$ 0.0577	16.86 $\pm$ 0.321	68.96 $\pm$ 0.057
	120	62.5 $\pm$ 0.360	19.53 $\pm$ 0.306	17.8 $\pm$ 0.264	80.3 $\pm$ 0.346
	150	73.1 $\pm$ 0.152	8.16 $\pm$ 0.305	18.66 $\pm$ 0.152	91.83 $\pm$ 0.305
	180	79.1 $\pm$ 0.70	0.4 $\pm$ 0.2	20.5 $\pm$ 0.5	16.0 $\pm$ 0.781
Al ₂ O ₃	30	9.23 $\pm$ 0.251	85.26 $\pm$ 0.25	5.5 $\pm$ 0.500	14.73 $\pm$ 0.12
	60	31.43 $\pm$ 0.493	55.16 $\pm$ 0.115	13.4 $\pm$ 0.458	44.83 $\pm$ 0.115
	90	54.5 $\pm$ 0.500	29.56 $\pm$ 0.602	15.93 $\pm$ 0.115	70.43 $\pm$ 0.602
	120	62.23 $\pm$ 0.25	1.33 $\pm$ 0.72	36.43 $\pm$ 0.92	98.66 $\pm$ 1.171
SiO ₂ –Al ₂ O ₃	30	19.36 $\pm$ 0.321	70.2 $\pm$ 0.529	10.43 $\pm$ 0.208	29.8 $\pm$ 0.529
	60	34.16 $\pm$ 0.763	41.43 $\pm$ 0.404	21.16 $\pm$ 0.288	58.56 $\pm$ 0.404
	90	51.56 $\pm$ 0.404	22.2 $\pm$ 0.346	26.23 $\pm$ 0.251	77.8 $\pm$ 0.34
	120	57.2 $\pm$ 0.264	15.3 $\pm$ 0.20	27.5 $\pm$ 0.30	84.7 $\pm$ 0.2
	150	62.56 $\pm$ 0.057	8.16 $\pm$ 0.208	29.26 $\pm$ 0.251	91.83 $\pm$ 0.2081
	180	67.43 $\pm$ 0.208	0.533 $\pm$ 0.152	32.03 $\pm$ 0.057	99.46 $\pm$ 0.152

Table 7b

Comparison of the optimum parameters for using different catalysts for thermal catalytic pyrolysis of waste polyethylene.

Parameters	CaC ₂	SiO ₂	MgO	Al ₂ O ₃	SiO ₂ –Al ₂ O ₃	ZnO
Temperature (°C)	350	350	350	350	400	400
Catalyst (g)	1	1	1.5	1.5	2	2
Time (h)	1	3	2	2	3	3
Liquid (%)	69.73 $\pm$ 0.64	79.10 $\pm$ 0.71	66.23 $\pm$ 0.25	62.23 $\pm$ 0.25	67.43 $\pm$ 0.21	64.93 $\pm$ 0.98

Table 8

Fractions of different hydrocarbons (wt%) obtained after elution from column using different solvents.

Catalyst type	Hydrocarbon (wt%)		
	Aliphatic	Aromatic	Polar
SiO ₂	35	20	40
SiO ₂ –Al ₂ O ₃	40	30	30
ZnO	30	30	40
MgO			
Dark brown layer	25	35	35
Denser layer	15	15	65
CaC ₂	40	30	25
Al ₂ O ₃	30	25	40

of CaC₂ as catalyst it was observed that the derived oil has maximum fraction of non-polar (40%) followed by relatively non-polar (30%) and minimum polar fraction (25%).

When MgO was used as a catalyst, two distinct layers, one upper dark brown in color and lower denser white layer were observed. These layers were collected separately and each one was subjected to separation on silica column using the same eluting solvents. It was observed that the white denser layer was mainly of polar components (65%) with fraction of relatively non-polar (15%) and highly non-polar (15%). The upper dark brown layer was found to be an equimolar mixture of highly non-polar, non-polar and polar fractions.

Looking into the results it can be inferred that the oxide containing catalyst could be best suited for selective conversion into polar and aromatic products while CaC₂ could be adopted for selective conversion into aliphatic products.

3.5. Characterization of the derived oil from catalytic pyrolysis of polyethylene

The oil obtained from catalytic pyrolysis with different catalysts without any further process has been named as parent oil. To find

out if there is any further cracking, each parent oil has been re-fluxed with respective catalyst at optimum temperature for 2 h time. The fuel properties like density, specific gravity, API gravity, viscosity, kinematic viscosity, aniline point, flash point, Watson characterization constant, freezing point, calorific value, diesel index, sulfur and ASTM distillation for both the parent oil and re-fluxed oil in each case has been measured and are given in [Tables 9a and 9b](#). The density for the parent oil in all cases was about 0.78 g/cm³ and no change was observed for the respective refluxed oil. On the basis of the value of densities both parent and the refluxed oil can safely be classed as kerosene (densities 0.78–0.82 g/cm³).

Specific gravity values for the parent oil ranges from 0.77 to 0.79 in case of SiO₂, SiO₂–Al₂O₃ and ZnO as a catalyst. Again no change in specific gravity was observed when parent oil was re-fluxed with respective catalyst. Again these values fall in the range for kerosene (0.78).

The values for API gravity were found to be in the range of 48.75–55.08 for parent and refluxed oil. It was observed that API gravity has slightly decreased on refluxing in case of refluxing with SiO₂, SiO₂–Al₂O₃, ZnO and MgO, while an increase in API gravity value upon refluxing was observed only in case of CaC₂ and Al₂O₃ as a catalyst. The API gravity values also fall within the API gravity values for Kerosene (39.66–41.7). Viscosity and kinematic viscosity values both for parent and refluxed oil were measured and are in the range 1.233–1.399 centipoises and 1.211–1.836 mm²/s respectively. It was observed that there is slight increase both in viscosity and kinematic viscosity upon refluxing with SiO₂ and SiO₂–Al₂O₃, while a slight decrease in these values was observed when the oil was refluxed with ZnO as catalyst. Viscosity values are in agreement with the viscosity values for kerosene (0.9–1.5centipoise).

The values for aniline point were found to be in the range of 65–79 °C both for the parent and refluxed oil. A decrease in aniline point was observed on refluxing with SiO₂, while increase in the same was observed upon refluxing with SiO₂–Al₂O₃ and ZnO as catalyst. The values of flash point for parent and refluxed oil varies

Table 9a

Fuel properties of oil derived from catalytic pyrolysis of polyethylene sample.

S. no.	Parameters	Oil derived with ZnO catalyst	Refluxed with ZnO catalyst	Oil derived with MgO catalyst	Refluxed with MgO catalyst	Oil derived with CaC ₂ catalyst	Refluxed with CaC ₂ catalyst
1	Density (g/cm ³)	0.768	0.769	0.765	0.786	0.759	0.765
2	Specific gravity	0.769	0.769	0.767	0.788	0.767	0.766
3	API gravity	52.50	52.48	53.03	48.14	52.93	53.12
4	Viscosity (centipoise)	1.267	1.258	1.123	1.185	1.084	1.096
6	Kinematic viscosity (mm ² /s)	1.650	1.638	1.468	1.51	1.429	1.432
7	Aniline point (°C)	69	71	70	69	68	65
8	Flash point (°C)	50	52	56	58	43	44
9	Watson characterization constant	12.172	12.189	11.982	11.938	12.163	12.179
10	Freezing point	−16	−14	−18	−19	−18	−19
11	Gross calorific value (MJ/kg)	40.5	40.6	44.1	44.6	43.1	43.3
12	Net calorific value (MJ/kg)	39.1	40.0	42.3	42.8	41.2	41.5
13	Diesel index	82.005	83.865	83.74	75.19	81.719	79.144
14	Sulfur (%)	0.64	0.67	0.75	0.7	0.68	0.69

Table 9b

Fuel properties of oil derived from catalytic pyrolysis of polyethylene sample.

S. no.	Parameters	Oil derived with SiO ₂ catalyst	Refluxed with SiO ₂ catalyst	Oil derived with Al ₂ O ₃ catalyst	Refluxed with Al ₂ O ₃ catalyst	Oil derived with SiO ₂ –Al ₂ O ₃ catalyst	Refluxed with SiO ₂ –Al ₂ O ₃ catalyst
1	Density (g/cm ³)	0.777	0.784	0.761	0.757	0.774	0.776
2	Specific gravity	0.778	0.785	0.762	0.758	0.776	0.777
3	API gravity	50.35	48.75	54.10	55.08	50.94	50.63
4	Viscosity (centipoise)	1.399	1.439	0.922	1.011	1.233	1.388
6	Kinematic viscosity (mm ² /s)	1.801	1.836	1.211	1.336	1.592	1.790
7	Aniline point (°C)	79	74	65	60	70	77
8	Flash point (°C)	62	63	30	32	53	54
9	Watson characterization constant	10.67	10.57	12.04	12.17	12.12	12.15
10	Freezing point	−12	−12	−20	−20	−14	−13
11	Gross calorific value (MJ/kg)	44.3	44.6	45.0	45.5	42.3	42.6
12	Net calorific value (MJ/kg)	42.6	42.8	42.8	43.0	41.1	41.4
13	Diesel index	87.75	80.54	80.617	77.144	80.48	86.37
14	Sulfur (%)	0.64	0.61	0.55	0.52	0.52	0.50

Table 10a

ASTM Distillation of oil derived from catalytic pyrolysis of polyethylene sample.

S. no.	Parameters	Oil derived with ZnO catalyst	Refluxed with ZnO catalyst	Oil derived with MgO catalyst	Refluxed with MgO catalyst	Oil derived with CaC ₂ catalyst	Refluxed with CaC ₂ catalyst
S. no.	ASTM distillation	Temp. (°C)	Temp. (°C)	Temp. (°C)	Temp. (°C)	Temp. (°C)	Temp. (°C)
1	Dew point	86	90	94	108	80	74
2	10 ml	118	120	102	136	117	122
3	20 ml	142	148	112	156	127	138
4	30 ml	172	168	122	164	150	156
5	40 ml	170	174	130	178	170	176
6	50 ml	186	188	132	190	178	178
7	60 ml	192	195	184	196	200	182
8	70 ml	201	202	196	204	205	200
9	80 ml	208	210	202	205	209	205
10	90 ml	212	216	208	209	213	209
11	100 ml	220	222	216	212	217	214

from 30 °C to 63 °C. It was observed that in all cases after refluxing the parent oil with catalyst, there was an increase in the flash point. It was further observed that flash point values were higher 52–63 °C, when SiO₂, SiO₂–Al₂O₃, ZnO and MgO were used as a catalyst, while flash point values were found lower (30–44 °C) when CaC₂ and Al₂O₃ were used as a catalyst, for cracking and refluxing. The flash point values indicate that the oils obtained using CaC₂ and Al₂O₃ as a catalyst may be gasoline (20.8–38 °C), while the oil obtained using catalytic cracking with SiO₂–Al₂O₃, ZnO and MgO may be kerosene (50–55 °C).

The Watson characterization constant value varies from 10.57 to 2.19 for parent and refluxed oil. No appreciable change in these values was observed on further refluxing with catalyst. From the

value of Watson characterization constant the oil may be considered aromatic and naphthenic in nature.

The values of freezing point, gross calorific value and net calorific values (Tables 9a and 9b) varies in the range of −12 °C to −20 °C, 40.5–45.5 MJ/kg and 39.1–43 MJ/kg respectively. In case of freezing point it was observed that there was no change in the value upon refluxing using SiO₂ and Al₂O₃ as a catalyst. A decrease in freezing point was observed when SiO₂–Al₂O₃ and ZnO was used for refluxing, while increase in freezing point value was observed when refluxing was done with MgO and CaC₂. The values of freezing point are in agreement with freezing point value of kerosene.

In case of gross calorific value and net calorific value a slight increase in the value was observed in all cases upon refluxing. In all

Table 10b

ASTM Distillation of oil derived from catalytic pyrolysis of polyethylene sample.

S. no.	Parameters	Oil derived with SiO ₂ catalyst Temp. (°C)	Refluxed with SiO ₂ catalyst Temp. (°C)	Oil derived with Al ₂ O ₃ catalyst Temp. (°C)	Refluxed with Al ₂ O ₃ catalyst Temp. (°C)	Oil derived with SiO ₂ –Al ₂ O ₃ catalyst Temp. (°C)	Refluxed with SiO ₂ –Al ₂ O ₃ catalyst Temp. (°C)
S. no.	ASTM distillation						
1	Dew point	104	130	68	82	92	98
2	10 ml	144	162	90	108	116	138
3	20 ml	174	186	105	118	154	170
4	30 ml	184	184	132	142	170	180
5	40 ml	199	192	152	160	180	196
6	50 ml	208	198	164	168	202	200
7	60 ml	216	204	170	172	205	202
8	70 ml	220	210	176	178	208	208
9	80 ml	221	216	184	188	212	214
10	90 ml	225	222	190	192	216	216
11	100 ml	228	226	198	204	220	218

Table 10c

Standards parameter of gasoline, diesel and kerosene oil.

S. no.	Parameters	Gasoline	Diesel	Kerosene
1	Density (g/ml)	0.736/0.725	0.834	0.780–0.82
2	Specific gravity	0.70	0.85	0.78
3	API gravity	65	23–30	41.7–39.66
4	Viscosity (centipoise)	0.7750–0.8394	2.0–4.5	0.9–1.5
5	Kinematic viscosity (mm ² /s)	5.0	3.77–5.0	2.2
6	Aniline point (°C)	65	71	62
7	Flash point (°C)	37.8–38	55–60	50–55
8	Watson characterization constant	12.45	11.28	12.126
9	Freezing point (°C)	–58	–54	–
10	Diesel index	83.44	54	59.88
11	Bromine number	60	1–10	
12	Gross calorific value (MJ/kg)	45.6	43.5–55.7	46.5
13	Sulfur (%)		0.70	
14	Net calorific value (MJ/kg)	42.8	41.1–42.9	43.5
S. no.	ASTM distillation	Temp. (°C)	Temp. (°C)	Temp. (°C)
0	Dew point	49	120	110
1	10 ml	69	225	178
2	20 ml	78	238	180
3	30 ml	82	252	180
4	40 ml	86	272	182
5	50 ml	90	281	190
6	60 ml	94	290	192
7	70 ml	98	299	198
8	80 ml	102	300	200
9	90 ml	106	301	210
10	100 ml	112	305	220

cases, these values are in agreement with the gross calorific and net calorific values of diesel that is 43.5 MJ/kg and 41.1 MJ/kg respectively. The value for diesel index both for parent and refluxed oil vary in the range of 75.19–87.75. It was observed that these values decrease in case of refluxing with SiO₂, MgO, CaC₂ and Al₂O₃, while an increase in these values are observed upon refluxing with SiO₂–Al₂O₃ and ZnO as catalyst. The result of the diesel index lie in the range of gasoline (83.44). Sulphur contents for parent and refluxed oil vary in the range of 0.50–0.75%. These values are in agreement to the sulfur contents for diesel and gasoline (0.7%). The results for ASTM distillation are given in Tables 10a and 10b. It can be seen that the initial boiling point in each case that it is far beyond the initial point of gasoline which is 37–40 °C in summer grade gasoline oil. In some cases, the dew point is very high which indicates that the oil under study mostly lie with in kerosene range. The final boiling point in all the cases is also in kerosene oil range (250 °C) (see Table 10c).

The dew point obtained in case of crackates with MgO as a catalyst was found to be 90 °C. The final boiling was found to be

216 °C. The liquid obtained in case of oil was refluxed with MgO at 350 °C for time duration of 2 h in order to get more rectification. The distillation behavior of such liquid was also studied. It is evident from the data obtained that the initial boiling point was further delayed and was found to be 108 °C as compared to that liquid obtained with MgO without reflux. It appears that during reflux some of the light fractions got converted into gaseous products due to further cracking.

The final boiling point was not altered which is an indicative of the fact that the lower boiling point constituents remain intact during heating under reflux. The liquid obtained with CaC₂ was also studied. The distillation data is given in the same table. It can be seen that the dew point is lesser than the one observed in the case of MgO as a catalyst only. This shows that the catalytic activity of this catalyst might have cleaved heavy molecular weight configuration into light oil which cause initial boiling point to occur at temperature less than the one determined in the case of the MgO. The final boiling point observed was 217 °C. In case of Al₂O₃ both for parent and refluxed oil the dew point was found to be 68 °C and 84 °C and final boiling point was 198 °C and 204 °C respectively. Which again verify our above mentioned explanation. Similarly incase of SiO₂, mixture of SiO₂–Al₂O₃ and ZnO both for parent and refluxed oil the dew point varies in the range of 86–104 °C and the final boiling point lie in the range of 218–228 °C, which again is in the range of kerosene oil.

3.6. Separation of the derived oil by fractional distillation and investigation of its physical parameters

The oil derived from catalytic pyrolysis of polyethylene using different catalysts were separated using fractional distillation at various temperatures ranging from 100 °C to 300 °C. Sufficient fraction was collected for its complete physical and chemical characterization. The results of fractionation and its physical parameters investigated are given in Tables 11 and 12.

It was observed from fractionation studies that when MgO is used as a catalyst, the derived oil contain predominantly lighter fraction, composed of volatile fraction. While in case of CaC₂ and mixture of SiO₂ and Al₂O₃ mostly semi volatile fractions are obtained. In case of Al₂O₃, ZnO and SiO₂ as a catalyst, the oil contained mainly heavy nonvolatile fractions.

All physical parameters like density, refractive index, refractive intercept, refractive index parameter and specific refraction for the derived oil were measured and are given in Tables 11 and 12.

It can be seen from the results that the densities and refractive indexes of oil increased from 0.510 to 0.956 g/cm³ and 1.330 to 1.516 respectively using catalysts like ZnO, MgO, CaC₂, SiO₂, Al₂O and mixture of SiO₂ and Al₂O₃. There is pronounced increase in refractive indexes and densities with increasing boiling point of

Table 11

Physical parameters of various fractions collected at different temperatures from fractional distillation of oil.

Type of catalysts	Temp. (°C)	%	Density (g/ml)	Refractive index (η)	Refractive intercept (RI)	Refractive index Parameter (I)	Specific refraction (γD)
ZnO	100	11.0	0.566	1.329	1.046	0.2034	0.102
	150	6.0	0.568	1.419	1.135	0.253	0.126
	180	24	0.866	1.433	0.991	0.260	0.130
	200	20	0.872	1.445	1.009	0.266	0.133
	250	10	0.892	1.446	1.000	0.267	0.133
	300	14	0.899	1.453	1.003	0.270	0.135
	Parent	9.0	0.916	1.435	0.998	0.272	0.136
	Residue	2.0	0.768	1.456	1.456	0.272	0.136
MgO							
Upper black light layer							
	100	7	0.740	1.410	1.040	0.248	0.127
	150	10	0.800	1.415	1.015	0.250	0.125
	180	6	0.840	1.425	1.005	0.256	0.128
	200	6	0.890	1.427	0.982	0.257	0.128
	250	12	0.895	1.438	0.991	0.263	0.131
	300	16	0.897	1.444	0.995	0.266	0.133
	Residue	22	0.901	1.4520	1.002	0.269	0.135
	Parent	100	0.899	1.439	0.990	0.263	0.132
Lower white denser layer							
	Parent	100	0.996	1.332	0.834	0.207	0.103
	100	83	0.965	1.330	0.848	0.204	0.102
	Residue	16.6	0.979	1.335	0.845	0.207	0.103
CaC ₂							
	170	15	0.812	1.416	1.010	0.250	0.125
	250	40	0.898	1.442	0.993	0.265	0.132
	300	13.33	0.899	1.446	0.996	0.266	0.133
	Residue	30	0.921	1.456	0.996	0.272	0.136
	Parent	25	0.900	1.441	0.991	0.264	0.132

Table 12

Physical parameters of various fractions collected at different temperatures from fractional distillation of oil.

Type of catalyst	Temp. (°C)	%	Density (g/cm ³)	Refractive index (η)	Refractive intercept (RI)	Refractive index Parameter (I)	Specific refraction (γD)
SiO ₂	100	5.5	0.510	1.404	1.149	0.244	0.122
	150	7.7	0.734	1.428	1.061	0.257	0.129
	180	10	0.737	1.435	1.066	0.261	0.130
	200	12.4	0.741	1.439	1.068	0.263	0.131
	250	16.6	0.764	1.444	1.061	0.265	0.133
	300	28.8	0.783	1.447	1.056	0.267	0.134
	Residue	35	0.793	1.462	1.065	0.275	0.138
	Parent	100	0.777	1.441	1.053	0.264	0.132
Al ₂ O ₃	100	4.42	0.722	1.452	1.091	0.269	0.135
	150	8.2	0.829	1.452	1.038	0.269	0.135
	180	16.8	0.829	1.465	1.050	0.276	0.138
	200	6.3	0.832	1.483	1.067	0.286	0.143
	250	22.1	0.855	1.499	1.072	0.294	0.147
	300	20.52	0.956	1.516	1.038	0.302	0.151
	Residue	13.68	0.962	1.556	1.075	0.321	0.161
	Parent	100	0.761	1.433	1.052	0.259	0.129
SiO ₂ –Al ₂ O ₃	100	11.0	0.710	1.406	1.051	0.245	0.123
	150	10.3	0.756	1.429	1.050	0.258	0.129
	180	8.5	0.803	1.452	1.049	0.269	0.135
	200	18.2	0.817	1.457	1.048	0.272	0.136
	250	15.8	0.848	1.472	1.048	0.280	0.140
	300	12.2	0.890	1.495	1.050	0.292	0.146
	Residue	18.3	0.921	1.562	1.102	0.324	0.162
	Parent	100	0.774	1.437	1.0499	0.262	0.131

the fraction. At lower temperature low molecular weight hydrocarbons are present and with increase in boiling point the molecular weight of the compounds are also increased. Low values of refractive indices are associated with paraffins and higher values with aromatic and intermediate with naphthenic hydrocarbons. The nature of the hydrocarbon group present in each fraction was further identified from the refractivity intercept (RI) and it gives specific value for each hydrocarbon group.

The RI value of paraffins lies in the range of 1.048–1.05, for naphthenic 1.03–1.046 and for aromatics 1.07–1.105. The RI values

for fractions obtained at 100–150 °C using MgO as a catalyst and for fraction obtained at 300 °C using Al₂O₃ as a catalyst lie in the range of naphthenic hydrocarbons, while in case of mixture of SiO₂ and Al₂O₃ all fractions obtained at 100–300 °C gives a value in the range of paraffinic hydrocarbons.

In case of Al₂O₃ at 100 °C and SiO₂ as a catalyst at 100 °C and 250 °C the RI values for fractions obtained lie in the range of aromatic hydrocarbons.

Olefinic hydrocarbons were determined in each fraction of liquid by bromine number. The results are given in Table 13 and

Table 13

Quantitative determination of bromine number, total phenol and carbonyl compounds in the fractions collected at different temperatures of oil.

Type of catalyst	Temp. (°C)	Bromine Number (g/ml)	Phenol (µg/ml)	Carbonyl (µg/ml)
ZnO	100	0.2	20	19
	150	0.3	117	36
	180	1.2	123	33
	200	5.3	45	29
	250	6.5	35	24
	300	7.3	92	18
	Residue	10.6	02	16
	Parent	11.5	48	169
MgO Upper black light layer	100	0.0	87	15
	150	0.1	99	35
	180	0.3	74	28
	200	3.8	62	25
	250	5.2	44	23
	300	6.2	88	16
	Residue	7.6	10	08
	Parent	9.2	120	147
Lower white denser layer	100	–	132	85
	Residue	–	87	25
	Parent	–	99	48
CaC ₂	170	1.9	2400	35
	250	6.5	1620	48
	300	7.3	4450	20
	Residue	10.6	450	19
	Parent	11.5	8920	123

Table 14

Quantitative determination of bromine number, total phenol and carbonyl compounds in the fractions collected at different temperatures.

Type of catalysts	Temp. (°C)	Bromine Number (g/ml)	Phenol (µg/ml)	Carbonyl (µg/ml)
SiO ₂	100	0.0	46.7	12
	150	0.1	96	29
	180	0.3	59	25
	200	0.7	41	23
	250	5.2	33	19
	300	7.2	45	13
	Residue	9.5	2.9	08
	Parent	10.2	48	131
Al ₂ O ₃	100	0.1	78	15
	150	0.2	76	27
	180	3.6	199	30
	200	3.8	131	28
	250	5.6	150	14
	300	7.5	224	12
	Residue	10.5	17	05
	Parent	11.9	77	142
SiO ₂ –Al ₂ O ₃	100	0.2	231	19
	150	0.4	104	23
	180	1.7	123	29
	200	5.6	260	21
	250	7.5	126	29
	300	8.6	145	19
	Residue	11.2	01	12
	Parent	12.8	167	169

14. As can be seen from the results that the bromine number ranges from 0.1 g/ml to 5.2 g/ml in the fractions from 100 °C to 250 °C, with MgO, Al₂O₃, ZnO, SiO₂–Al₂O₃, SiO₂, and CaC₂ as catalyst, which shows small concentration of olefinic hydrocarbons. With further increase of temperature there is an increase in the bromine number from 5.2 g/ml to 9.5 g/ml shows the presence of sufficient amount of olefinic hydrocarbons.

The oxygenated compounds were determined quantitatively in terms of total phenols and carbonyls. The results are given in Table 13 and 14. The fractions with CaC₂ catalyst show relatively higher concentration of total phenols as compared to other.

The presence of small concentration of phenols and carbonyls indicates that it may be due to the pyrolysis, cyclization and inter-action of cracking product with atmospheric oxygen at high temperature in the presence of catalyst.

3.7. Infrared analysis of the derived oil from catalytic pyrolysis of polyethylene

The derived oil was expected to be a complex mixture of hydrocarbons with major aliphatic fraction. It was expected that no complete separation could be achieved either with fractional distillation or column separation and as a result IR studies would be inconclusive. However IR spectra was recorded for parent oil, refluxed oil and fractions obtained by column separation and fractional distillation.

Looking into the IR spectra recorded, it was observed that absorption bands were observed for alkanes, alkenes and smaller concentration of aromatic and oxygenated compounds.

4. Conclusion

A wide range of acidic and basic catalysts like silica, calcium carbide, alumina, magnesium oxide, zinc oxide and homogeneous mixture of silica and alumina were used for conversion of waste polyethylene. Though CaC₂ was better on the basis of reaction time, however the efficiency of conversion into liquid for SiO₂ was found to be maximum in case of LDPE at optimum conditions. These two catalysts could be picked up as suitable catalysts for catalytic pyrolysis of polyethylene.

The results of the column separation using different solvents indicate that the oxide containing catalyst could be best suited for selective conversion into polar and aromatic products while CaC₂ could be adopted for selective conversion into aliphatic products.

From the physical tests it was observed that the results for the liquid fractions are comparable with the standard results of physical tests for gasoline, kerosene and diesel fuel oil. From the Bromine water and KMnO₄ tests it was observed that liquid obtained is a mixture of olefin and aromatic hydrocarbons. This was further confirmed by Bromine number tests. The values of which lie in the range of 0.1–12.8 g/ml for LDPE which fall in the range for olefin mixture. Phenol and carbonyl contents were quantified using UV/Visible spectroscopy and the values lie in the range of 1–8920 µg/ml and 5–169 µg/ml for LDPE for both phenols and carbonyls respectively. The components of different hydrocarbons in the oil mixture were separated by using column chromatography and fractional distillation followed by characterization with FT-IR spectroscopy.

The interpretation of FT-IR spectra show that catalytic pyrolysis of LDPE leads to the formation of a complex mixture of alkanes, alkenes, carbonyl group containing compounds like aldehydes, ketones, aromatic compounds and substituted aromatic compounds like phenols.

It could be concluded, that catalytic pyrolysis of LDPE leads to valuable resource recovery and reduction of waste problem.

References

- [1] Amrousi FAA. An unconventional cracking method for hydrocarbon compounds and their derivatives: 1. Liquefaction of polyolefins. *Polym Degrad Stabil* 2000;70:365–71.
- [2] Kiran N, Ekinici E, Snape CE. Recycling of plastic wastes via pyrolysis. *Res Conserv Recycl* 2000;29:273–83.

- [3] Arena U, Mastellone ML. Defluidization phenomena during the pyrolysis of two plastic wastes. *J Chem Eng Sci* 2000;55:2849–60.
- [4] Buekens AG, Huang H. Catalytic plastics cracking for recovery of gasoline-range hydrocarbons from municipal plastic wastes. *Polym Degrad Stabil* 2000;70:365–71.
- [5] Fang Y, Zhan M, Wang Y. The status of recycling of waste rubber. *J Mater Des* 2001;22:123–8.
- [6] Pakdel H, Pante DM, Roy C. Production of *dl*-limonene by vacuum pyrolysis of used tyres. *J Anal Appl Pyrol* 2001;57:91–107.
- [7] Gonzalez JF, Encinar JM, Canito JL, Rodriguez JJ. Pyrolysis of automobile tyre waste, influence of operating variables and kinetics study. *J Anal Appl Pyrol* 2001;58:667–83.
- [8] Kaminsky W, Mennerich C. Pyrolysis of synthetic tyre rubber in a fluidized – bed reactor to yield 1, 3-butadiene, styrene and carbon black. *J Anal Appl Pyrol* 2001;58:803–11.
- [9] Ballice L. A kinetic approach to the temperature-programmed pyrolysis of low- and high-density polyethylene in a fixed bed reactor, determination of kinetic parameters for the evolution of *n*-paraffins and 1-olefins. *Fuel* 2001;80:1923–35.
- [10] Chen JH, Chen KS, Tong LY. On the pyrolysis kinetics of scrap automotive tyres. *J Hazard Mater* 2001;84:43–55.
- [11] Jakaband E, Blazso M. The effect of carbon black on the thermal decomposition of vinyl polymers. *J Anal Appl Pyrol* 2002;64:263–77.
- [12] Murata K, Hirano Y, Sakata Y, Uddin MA. Basic study on a continuous flow reactor for thermal degradation of polymers. *J Anal Appl Pyrol* 2002;65:71–90.
- [13] Ucar S, Karagoz S, Karayildirim T, Yanik J. Conversion of polymers to fuels in a refinery stream. *Polym Degrad Stabil* 2002;75:161–71.
- [14] Leung DYC, Yin XL, Zhao ZL, Xu BY, Chen Y. Pyrolysis of tyre powder: influence of operation variables on the composition and yields of gaseous product. *Fuel Process Technol* 2002;79:141–55.
- [15] Faravelli T, Bozzano G, Colombo M, Ranzi E, Dente M. Kinetic modeling of the thermal degradation of polyethylene and polystyrene mixtures. *J Anal Appl Pyrol* 2003;70:761–77.
- [16] Bhaskar T, Uddin MA, Murai K, Kaneko J, Hamano K, Kusaba T, et al. Comparison of thermal degradation products from real municipal waste plastic and model mixed plastics. *J Anal Appl Pyrol* 2003;70:579–87.
- [17] Ding W, Liang J, Anderson LL. Polyethylene and commingled post-consumer plastic waste. *J Appl Catal* 2004;B49:257–65.
- [18] Font R, Aracil I, Fullana A, Conesa JA. Semivolatile and volatile compounds in combustion of polyethylene. *J Chem* 2004;57:615–27.
- [19] Murata K, Sato K, Sakata Y. Effect of pressure on thermal degradation of polyethylene. *J Anal Appl Pyrol* 2004;71:569–89.
- [20] Koc A, Bilgesu A, Alibeyli R, Kocak MC. Factorial experimental design for oxidative thermal decomposition of low-density polyethylene waste. *J Anal Appl Pyrol* 2004;72:309–15.
- [21] Park DW, Hwang EY, Kim JR, Choi JK, Kim YA, Woo HC. Catalytic degradation of polyethylene over solid acid catalysts. *J Appl Catal* 2000;A192:273–80.
- [22] Lee KH, Shin DH, Seo YH. Liquid-phase catalytic degradation of mixtures of waste high-density polyethylene and polystyrene over spent FCC catalyst and effect of mixing proportions of reactants. *J Anal Appl Pyrol* 2001;58:789–801.
- [23] Serrano DP, Aguado J, Escola JM, Garagorri E. Performance of a continuous screw kiln reactor for the thermal and catalytic conversion of polyethylene – lubricating oil base mixtures. *J Anal Appl Pyrol* 2001;59:127–42.
- [24] Schirmer J, Kim JS, Klemm E. Catalytic degradation of polyethylene using thermal gravimetric analysis and a cycled-spheres-reactor. *J Anal Appl Pyrol* 2001;60:205–17.
- [25] You YS, Kim JH, Seo G. Liquid-phase catalytic degradation of polyethylene wax over silica-modified zeolite catalysts. *Polym Degrad Stabil* 2001;72:329–36.
- [26] Lingaiah N, Uddin MA, Muto A, Imai T, Sakata Y. Removal of organic chlorine compounds by catalytic dehydrochlorination for the refinement of municipal waste plastic derived oil. *Fuel* 2001;80:1901–5.
- [27] Masuda T, Kushino T, Matsuda T, Mukai SR, Hashimoto K, Yoshida SI. Chemical recycling of mixture of waste plastics using a new reactor system with stirred heat medium particles in steam atmosphere. *Chem Eng* 2001;82:173–81.
- [28] Aguado J, Serrano DP, Escola JM, Garagorri E. Catalytic conversion of low-density polyethylene using a continuous screw kiln reactor. *J Therm Acta* 2002;55:392–3.
- [29] Marcilla A, Gomez A, Garcia AN, Olaya MM. Kinetic study of the catalytic decomposition of different commercial polyethylenes over an MCM-41 catalyst. *J Anal Appl Pyrol* 2002;64:85–101.
- [30] Sakata Y, Uddin MA, Muto A, Kanada Y, Koizumi K, Murata K. Catalytic degradation of polyethylene into fuel oil over mesoporous silica (KFS-16) catalyst. *J Anal Appl Pyrol* 2002;65:185–95.
- [31] Jalil PA. Investigations on polyethylene degradation into fuel oil over tungstophosphoric acid supported on MCM-41 mesoporous silica. *J Anal Appl Pyrol* 2002;65:196–225.
- [32] Fernandes GJT, Fernandes VJ, Araujo AS. Catalytic degradation of polyethylene over SAPO-37 molecular sieve. *J Catal Today* 2002;75:233–8.
- [33] Karagoz S, Yanik J, Ucar S, Saglam M, Song C. Catalytic and thermal degradation of high-density polyethylene in vacuum gas oil over non-acidic and acidic catalysts. *J Catal Today* 2002;75:257–62.
- [34] Ali S, Garforth AA, Harris DH, Rawlence DJ, Uemichi Y. Polymer waste recycling over used catalysts. *J Catal Today* 2002;75:247–55.
- [35] Sakata Y, Uddin MA, Muto A. Degradation of polyethylene and polypropylene into fuel oil by using solid acid and non-acid catalysts. *J Appl Catal B, Environ* 2003;44:95–105.
- [36] Serrano DP, Aguado J, Escola JM, Rodríguez JM, Morselli L, Orsi R. Thermal and catalytic cracking of a LDPE-EVA copolymer mixture. *J Appl Catal* 2003;A24:51–62.
- [37] Gao Z, Amasaki I, Nakada M. A thermogravimetric study on thermal degradation of polyethylene. *J Anal Appl Pyrol* 2003;67:1–9.
- [38] Seddegi ZS, Budrthumal U, Arfaj AA, Amer AM, Barri SAI. Catalytic cracking of polyethylene over all-silica MCM-41 molecular sieve. *J Anal Appl Pyrol* 2003;68–69:481–94.
- [39] Zhou Q, Zheng L, Wang YZ, Zhao GM, Wang B. Catalytic degradation of low-density polyethylene and polypropylene using modified ZSM-5 zeolites. *J Anal Appl Pyrol* 2003;70:383–98.
- [40] Karagoz S, Karayildirim T, Ucar S, Yuksel M, Yanik J. Liquefaction of municipal waste plastics in VGO over acidic and non-acidic catalysts. *Fuel* 2003;82:415–23.
- [41] Miskolczi N, Bartha L, Deak G, Jover B, Kallo D. Thermal and thermo-catalytic degradation of high-density polyethylene waste. *J Anal Appl Pyrol* 2004;72:235–42.
- [42] Ali MF, Siddiqui MN. Thermal and catalytic decomposition behavior of PVC mixed plastic waste with petroleum residue. *J Pyrol* 2004;74:282–9.
- [43] Seo YH, Lee KH, Shin DH. Investigation of catalytic degradation of high-density polyethylene by hydrocarbon group type analysis. *Polym Degrad Stabil* 2004;83:267–79.
- [44] Bagri R, Williams PT. Catalytic pyrolysis of polyethylene. *Polym Degrad Stabil* 2004;84:123–7.
- [45] Uddin MA, Sakata Y, Muto A, Shiraga Y, Koizumi K, Kanada Y, et al. Catalytic degradation of polyethylene and polypropylene into liquid hydrocarbons with mesoporous silica. *Polym Degrad Stabil* 2004;84:459–67.
- [46] Serrano DP, Aguado J, Escola JM, Garagorri E, Rodríguez JM, Morselli L, et al. Feedstock recycling of agriculture plastic film wastes by catalytic cracking. *Polym Degrad Stabil* 2004;84:493–7.
- [47] Gobin K, Manos G. Polymer degradation to fuels over microporous catalysts as a novel tertiary plastic recycling method. *Polym Degrad Stabil* 2004;86:121–8.
- [48] Gobin K, Manos G. Thermogravimetric study of polymer catalytic degradation over micro porous materials. *Polym Degrad Stabil* 2004;86:225–31.
- [49] Miskolczi N, Bartha L, Deak G, Jover B. Thermal degradation of municipal plastic waste for production of fuel-like hydrocarbons. *Polym Degrad Stabil* 2004;86:357–66.
- [50] Garcia RA, Serrano DP, Otero D. Catalytic cracking of HDPE over hybrid zeolitic-mesoporous materials. *J Anal Appl Pyrol* 2005;73:314–22.
- [51] Hajekova E, Bajus M. Recycling of low-density polyethylene and polypropylene via coprolysis of polyalkene oil/waxes with naphtha product distribution and coke formation. *J Anal Appl Pyrol* 2005;74:270–81.
- [52] You YS, Kim JH, Seo G. Liquid-phase catalytic degradation of polyethylene wax over MFI zeolites with different particles sizes. *Fuel Process Technol* 2005;86:1265–78.
- [53] Brebu M, Bhaskar T, Murai K, Muto A, Sakata Y, Uddin MA. Thermal degradation of PE and PS mixed with ABS-Br and debromination of pyrolysis oil by Fe- and Ca-based catalysts. *Polym Degrad Stabil* 2005;87:117–20.
- [54] Akpanudoh NS, Gobin K, Manos G. Catalytic degradation of plastic waste to liquid fuel over commercial cracking catalysts and effect of polymer to catalyst ratio/acidity content. *J Mole Catal* 2005;A235:67–73.
- [55] Aguado J, Serrano DP, Miguel GS, Castro MC, Madrid S. Feedstock recycling of polyethylene in a two-step thermo-catalytic reaction system. *J Anal Appl Pyrol* 2007;79:415–23.
- [56] Shah J, Jan MR, Mabood F. Catalytic conversion of waste tyres into valuable liquid fuels. *J Polym Environ* 2007;15(3):207–11.
- [57] Shah J, Jan MR, Mabood F. Conversion of waste tires into liquid hydrocarbons via base catalysis. *Iran J Chem Chem Eng* 2008;27(3).
- [58] Shah J, Jan MR, Mabood F. Recovery of value-added product from the catalytic pyrolysis of waste tyre. *Energy Convers Manage* 2009;50(4):991–4.